

NICHOLAS WOOD

Staff Software Engineer | AI & Full Stack Development

✉ knic43220@gmail.com ☎ 9177950554 📍 Regina Rd - Hillsboro - MO-63050

SUMMARY

- Staff Software Engineer with over nine years of experience designing large-scale, **AI-driven platforms** across healthcare and enterprise domains.
- Design event-driven, real-time and batch pipelines (**Kafka/Kinesis/Spark/Airflow/dbt**) with robust governance(**SOC2/HIPAA**), and productionize **ML/LLM** features with lineage, quality, and cost/latency guardrails
- Hands-on experience integrating **LLM** orchestration frameworks like **LangChain**, **LangGraph**, and **MCP** into production workflows to automate operations and improve efficiency.
- Strong background in **EHR** and **FHIR** integrations, **HIPAA-compliant** architectures, and secure cloud infrastructure for AI and data-intensive applications.
- Comfortable owning the full stack from ingestion to analytics: schema design (**Postgres/Redshift/Snowflake**), lakehouse (**S3/Delta/Glue**), orchestration (**Airflow**, **Step Functions**), observability (**OpenTelemetry/Prometheus**),and **CI/CD** (**Docker/GitHub Actions/Terraform/CDK**).
- Known for leading **cross-functional teams**, **mentoring** engineers, and delivering reliable, scalable software that bridges modern AI with real- world business impact.

EDUCATION

Missouri University of Science and Technology, 07/2014 – 07/2016 | Rolla, MO
Master's in Computer Science Degree

Missouri University of Science and Technology, 09/2010 – 06/2014 | Rolla, MO
Bachelor's in Computer Science Degree

EXPERIENCE

Notable  , Staff Software Engineer 08/2022 – 11/2025 | San Mateo, CA
An AI platform for healthcare operations that automates patient-access, clinical, and revenue-cycle workflows across major EHRs. Think agentic workflows + low/no-code configuration + deep EHR/payer integrations.

- Spearheaded the design and scaling of Notable's AI automation platform, integrating event-driven **FastAPI**, **Node.js**, and **Golang** microservices, **LLM** orchestration with **LangGraph**, vector search (**pgvector/Pinecone**), and streaming inference endpoints, facilitating over a million autonomous agent workflows daily across enterprise healthcare systems.

- Orchestrated the build-out of multi-tenant event-driven services in **FastAPI** and **Node/TypeScript**, establishing connections with major **EHR** systems (Epic, Cerner, Athena) through secure **FHIR** and **HL7** integrations.
- Engineered internal **MCP-based connectors** enabling AI agents to directly interact with systems such as Outlook, Salesforce, CRMs, and insurance APIs without human intervention.
- Collaborated closely with the **AI/ML** team to deploy **LangChain** and **LangGraph**-based orchestration flows, enabling conversational and reasoning agents that summarize claims, verify benefits, and assist with clinical documentation.
- Revamped **vector-search pipelines** (pgvector + Pinecone) for matching unstructured documents and automating prior-authorization lookups across patient records.
- Designed secure tokenization and data isolation strategies for **PHI** and user-level audit trails to uphold full **HIPAA** and **SOC 2 compliance**.
- Established streaming and observability pipelines (**Kafka**, **Postgres**, **S3**, **CloudWatch**) to monitor automation latency, error patterns, and ROI metrics across thousands of care sites.
- Managed **AWS**-based infrastructure supporting **LLM** orchestration, vector search, and event-driven APIs, optimizing the use of **ECS Fargate**, **Lambda**, **S3**, **CloudWatch**, and **RDS** (Postgres) for scalable, HIPAA-compliant operations.
- Conducted periodic security and compliance audits (**IAM** policies, encryption at rest, network isolation) to maintain SOC 2 and HIPAA certification standards.
- Developed real-time data interfaces for **AI agents** and automation workflows utilizing **SSE**, **WebSockets**, and **React Query**, ensuring instant feedback loops between clinicians and back-office systems.
- Partnered with product designers to create an AI activity dashboard visualizing automation states, token usage, and workflow metrics using **Recharts**, **D3.js**, and **Grafana** panels.
- Collaborated with clinicians and operations leads to translate manual workflows into AI-first agent flows, reducing repetitive administrative work by over 60% and expediting patient intake cycles.
- Mentored a small team of backend and **ML** engineers, guiding them on service boundaries, structured retries, and **CI/CD** practices with **AWS CDK**.
- Developed internal tools and dashboards in Looker and **Grafana** to monitor agent success rates, token usage, and cost metrics for LLM inference.
- Guided cross-functional engineers in test-driven development (**TDD**), scalable service design, and best practices for deploying reliable, production-grade AI systems.

BitsWits, Senior Data Engineer

06/2018 – 06/2022 | Huston, TX

Built large-scale, cloud-native data and event-driven platforms for enterprise and startup clients, with a heavy focus on streaming architectures, real-time analytics, and AWS-based operational data pipelines supporting web and mobile applications.

- Directed the end-to-end design and development of large-scale web and mobile applications for enterprise and startup clients, utilizing React, Next.js, Node.js, Golang, and Python (FastAPI) with PostgreSQL and Redis.
- Collaborated directly with product and client teams to transform requirements into scalable architecture diagrams, GraphQL APIs, and AWS-based deployment pipelines (ECS, Lambda, S3, CloudFront).

- Constructed event-driven microservices supporting analytics, notifications, and behavior tracking, driven by Kafka, Redis Streams, and GraphQL federation gateways.
- Engineered real-time communication layers (WebSockets, Server-Sent Events, and WebRTC) for live dashboards, chat, and telemetry, effectively reducing latency and enhancing engagement by over 10%.
- Unified front-end codebases by establishing a shared React/React Native component library and implementing monorepo tooling (Nx, Turborepo), decreasing I/J development cycles by over 25%.
- Instituted TypeScript, Jest, ESLint, and Cypress as engineering standards, promoting the adoption of TDD and uniform coding practices across multiple product teams.
- Implemented CI/CD pipelines with GitHub Actions and Docker, automating integration tests, staging previews, and zero-downtime deployments.
- Integrated third-party APIs (Stripe, Firebase, Twilio, Google Maps) and developed secure authentication flows using JWT, OAuth 2.0, and AWS Cognito for multi-tenant applications.
- Mentored engineers in clean coding practices, TDD workflows, peer review discipline, and observability best practices, enhancing performance and reliability across projects.
- Participated in architecture reviews, hiring panels, and internal governance, influencing BitsWits' engineering standards for scalability, testing, and security.

Oracle Health , *Data Enigineer*

01/2017 – 05/2018 | Kansas City, MO

Healthcare IT/EHR platforms powering patient scheduling, orders, documentation, and integrations. Focused heavily on health data interoperability, HL7/FHIR pipeline reliability, and compliant data platform engineering.

- Delivered features and fixes across Java/Spring services supported by Oracle Database (SQLPL/SQL), exposing REST endpoints utilized by patient scheduling, orders, and documentation modules.
- Developed HL7/FHIR adapters for internal services (ADT, orders, results), incorporating validation and error routing that reduced integration failure tickets by approximately 20%.
- Authored unit and integration tests with JUnit/Mockito and established lightweight contract tests for downstream consumers, improving coverage on a core service from approximately 5% to 70%.
- Assisted in migrating legacy SOAP endpoints to REST with clearer resource models, request validation, and standardized error responses.
- Contributed to CI/CD with Jenkins (later GitHub Enterprise), incorporating static analysis (SpotBugs/Checkstyle), test gates, and artifact versioning; effectively reducing release rollbacks during quarterly drops.
- Containerized development/test environments with Docker, creating reproducible local stacks, and documenting runbooks that accelerated onboarding for new engineers.
- Implemented PHI-safe audit logging, encryption-in-transit (mTLS), and role-based access checks, aligning services with HIPAA and internal compliance guidelines.
- Collaborated with analysts and QA to triage production issues, reproduce edge cases, and deliver hotfixes with clearly defined rollback plans and change records.

Oracle Health, *Engineer Intern*

07/2016 – 12/2016 | Kansas City, MO

SKILLS

AI/ML

LLM Development & Fine-Tuning, RAG, Langchain, Langgraph, Langbase, LangSmith, Hugging Face, Autogen, CrewAI, DSPy, Open AI, Llama 3, PyTorch, TensorFlow, Ray, MLflow, Pinecone, pgvector, FAISS, Multi agent system, Prompt Engineering & Evaluation, SageMaker, GCP Vertex AI, CUDA

Languages & Runtime

JavaScript (ES6+), TypeScript, Python, Node.js, Golang, Java (JVM), C# (.NET), C++, Rust, SQL, Bash/Shell/PowerShell, YAML/JSON/Protobuf, GraphQL, CUDA, Julia, WebAssembly(WASM), Ray/Dask, Hugging Face Inference/OpenAI/vLLM runtimes

DevOps

Terraform, AWS CDK, CloudFormation, Ansible, AWS (ECS, Fargate, Lambda, S3, RDS, CloudWatch, EventBridge), GCP & Azure, CI/CD, Github Actions, Jenkins, GitLab CI, Docker, Kubernetes, Prometheus/Grafana, ELK, OpenTelemetry, CloudWatch, SLO/SLA/SLI Design, VPC & Networking, VPN, Zero-Trust, Auto-Scaling

Data Engineering

Event-driven pipelines (Kafka, Redis Streams, SQS/SNS), ETL / ELT pipeline design & orchestration, Change Data Capture (CDC), replayable ingestion, Batch + streaming data processing, Data modeling (OLTP + analytical), PostgreSQL, Oracle SQL/PLSQL, Snowflake, BigQuery, Columnar & warehouse schemas, S3 / Lakehouse patterns, Kafka (producers, consumers, schema evolution),

Backend & API

Microservices & Distributed Systems, Rest, GraphQL, gRPC, Node.js (Express, NestJS), Python (Flask, FastAPI, Django), Golang(Gin, Fiber), .NET Core/Java Spring Boot, OAuth2, JWT, SSO, RBAC/ABAC, Prisma, TypeORM, SQLAlchemy, EF, Gorm, Redis, Kafka, RabbitMQ, AWS SQS, Websocket, WebRTC, SSE, Pub/Sub, AWS API gateway, NGINX, Kong, Cloudflare

Testing

TDD, BDD, Pytest/unittest (Python), Jest/Mocha/Vitest, xUnit/Nunit, Cypress, Selenium, Playwright, Model Evaluation Pipelines, Prompt Evaluation, Pandera, k6, Locust, JMeter, OWASP ZAP